

Junseo Kim

PHD STUDENT · ROBOTICS ENGINEER

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Summary

A robotics researcher object interaction using dexterous hand, and humanoid. Studying algorithms that are robust generalizable, better sim2real transfer, with the goal of achieving human-level dexterity for robot manipulation. Currently, a PhD Student at the KTH royal institute of Technology under supervision of Prof. Danica Kragic. Previously worked as a visiting research scholar at the BIG Lab at CMU, Pittsburgh, PA, United States.

Field of Interest

MY RESEARCH INTERESTS INCLUDE, BUT ARE NOT CONFINED TO:

- Dexterous Manipulation
- Humanoid whole-body control
- Motion Retargeting from MoCap, Video data
- Wireless signal processing for Robotics application
- Visual-Inertial Odometry

Education

* includes 18-months military service

KTH Royal Institute of Technology

Stockholm, Sweden

PH.D. IN COMPUTER SCIENCE

Aug. 2026 - Present

- Advised by Prof. Danica Kragic

Delft University of Technology (TU Delft)

Delft, Netherlands

M.S. IN ROBOTICS

Sept. 2024 - Jul. 2026

- Advised by Prof. Javier Alonso-Mora, Prof. Jean Oh (CMU)

Toronto Metropolitan University

Toronto, ON, Canada

B.ENG. IN MECHANICAL ENGINEERING

Sept. 2018 - Jun. 2024*

- Advised by Prof. Sajad Saeedi

Work Experience (Including Internship)

Visiting Research Scholar

Pittsburgh, PA, United States

BOT INTELLIGENCE GROUP, CMU

Aug. 2025 - Jun. 2026

- Advised by Prof. Jean Oh.
- Conducted research on video action model using kinematic rendering.

Publications

Underline denotes co-first authors or co-corresponding authors

INTERNATIONAL JOURNAL

1. Junseo Kim, Matthew Lisondra, Yeganeh Bahoo, and Sajad Saeedi, “Inverse k-visibility for RSSI-based Indoor geometric mapping”, *Autonomous Robots, Springer Nature*, vol. 50, no. 2, pp. 17, 2026, <https://doi.org/10.1007/s10514-026-10243-w>
2. Ishaan Mehta, Junseo Kim, Sharareh Taghipour, and Sajad Saeedi, “M³RS: Multi-robot, Multi-objective, and Multi-mode Routing and Scheduling”, *Journal of Field Robotics*, 0: 1–16, <https://doi.org/10.1002/rob.70288>
3. Matthew Lisondra, Junseo Kim, Glenn Takashi Shimoda, Kourosh Zareinia, and Sajad Saeedi, “TCB-VIO: Tightly-Coupled Focal-Plane Binary-Enhanced Visual Inertial Odometry”, *IEEE Robotics and Automation Letters*, vol. 10, no. 12, pp. 12341-12348, Dec. 2025, <https://doi.org/10.1109/LRA.2025.3619774>

INTERNATIONAL CONFERENCE

1. **Junseo Kim**, Guido Dumont, Xinyu Gao, Gang Chen, Holger Caesar, and Javier Alonso-Mora, “MobileOcc: A Human-Aware Semantic Occupancy Dataset for Mobile Robots,” in *European Conference on Computer Vision (ECCV)*, Malmö, Sweden, 2026. https://doi.org/10.1007/978-3-032-37013-6_36
2. Pablo Ortega-Kral, Eliot Xing, Arthur Buckner, Vernon Luk, **Junseo Kim**, Owen Kwon, Angchen Xie, Nikhil Sobanbabu, Yifu Yuan, Megan Lee, Deepam Ameria, Bhaswanth Ayapilla, Jaycie Bussel, Guanya Shi, Jonathan Francis, and Jean Oh, “RIO: Flexible Real-Time Robot I/O for Cross-Embodiment Robot Learning,” in *Robotics: Science and Systems (RSS)*, Sydney, Australia, 2026, <https://doi.org/10.15607/RSS.2026.XXII.164>
3. Huichan Seo, Sieun Choi, Minki Hong, Yi Zhou, **Junseo Kim**, Lukman Ismaila, Naome Etori, Mehul Agarwal, Zhixuan Liu, Jihie Kim, and Jean Oh, “Exposing Blindspots: Cultural Bias Evaluation in Generative Image Models,” in *Proceedings of IASEAI Conference, Paris, France, 2(1)*, 724–736, <https://ojs.aaai.org/index.php/IASEAI/article/view/43063>
4. **Junseo Kim**, Jill Aghyourli Zalat, Yeganeh Bahoo, and Sajad Saeedi, “Structure from WiFi (SfW): RSSI-Based Geometric Mapping of Indoor Environments,” in *American Control Conference (ACC)*, Toronto, ON, Canada, 2024, pp. 259-264, <https://doi.org/10.23919/ACC60939.2024.10644833>.
5. Matthew Lisondra, **Junseo Kim**, Riku Murai, Kourosh Zareinia, and Sajad Saeedi, “Visual Inertial Odometry using Focal Plane Binary Features (BIT-VIO),” in *IEEE International Conference on Robotics and Automation (ICRA)*, Yokohama, Japan, 2024, pp. 1661-1668, <https://doi.org/10.1109/ICRA57147.2024.10610838>

Honors & Awards

EXTERNAL AWARDS

2023 **Korean Canadian Scientists Scholarship Foundation Award**, AKCSE

Toronto

INTERNAL AWARDS

2025 **Justus and Louise van Effen Research Grant**, TU Delft

Delft

2024 **Justus and Louise van Effen Excellence Scholarship**, TU Delft

Delft

2024 **Dean's List**, TMU

Toronto

2023 **UIRO(Undergraduate Interdisciplinary Research Opportunities) Award**, TMU

Toronto

Academic Services

2026 **Reviewer**, ICRA, ECCV

2025 **Reviewer**, IROS